

Zijie Cai

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EDUCATION

University of Maryland, College Park

Aug 2024 – May 2025

Master of Science in Computer Science; GPA: 3.70 / 4.00

College Park, MD

University of Maryland, College Park

Aug 2020 – May 2024

Bachelor of Science in Computer Science – Machine Learning Specialization; GPA: 3.70 / 4.00

College Park, MD

- Minor in Mathematics; GPA: 3.72 / 4.00
- Dean's List (Multiple Semesters)
- Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Computational Imaging, Computational Biology, Algorithms, Data Structures, Parallel Computing, Applied Probability, Linear Algebra

RESEARCH EXPERIENCE

Intelligent Sensing Laboratory, University of Maryland | [Website](#)

College Park, MD

Research Assistant – Multimodal Foundation Models & Machine Perception

Apr 2023 – Present

- **Master's Paper:** *Underwater Monocular Metric Depth Estimation: Benchmarks and Fine-Tuning* [[arXiv](#)]
- Developed deep learning models for underwater depth estimation using aligned RGB and sonar data.
- Led experiments and benchmarking using synthetic and real-world RGB-D and acoustic sonar datasets.
- Presented weekly progress and co-authored publications under Prof. [Christopher Metzler](#).
- Collaborated on multi-modal pipelines for Autonomous Underwater Vehicle-based underwater perception.

WORK EXPERIENCE

Internal Drive Inc. | [Website](#)

College Park, MD

Program Lead – AI & Machine Learning

May 2025 – Present

- Delivered project-based instruction on **Python**, **machine learning**, and **prompt engineering**, teaching students to build AI tools using the **OpenAI API**, **Keras**, **NumPy**, and **Pandas**.
- Designed and led hands-on labs in **computer vision**, **natural language processing**, and **text/image/audio generation** with tools including **Teachable Machine**, and **TTS/STT APIs**.
- Mentored diverse student teams through full ML pipelines: data preprocessing, model development, API integration, debugging, and iterative refinement; while fostering technical communication and collaboration.

PROJECTS

PupCare (Beta) – Flutter/Dart, Firebase, Google Maps API | [Website](#)

- Created a cross-platform mobile app using **Flutter/Dart** with features including live **GPS walk tracking**, habit logging, and photo/video journaling backed by **Firebase Auth**, **Firestore**, and **Cloud Storage**.
- Implemented an interactive map with **Google Maps API**, custom markers, and a location-based stamp system to enhance discovery and generate personalized activity timelines for users.

UMD Class Planning Tool – JavaScript, HTML/CSS, Python, Web Scraping | [Website](#)

- Developed a responsive scheduling tool for **UMD students**, combining class times, commute routes, and instructor ratings into a unified **JavaScript** interface to support clearer decision-making.
- Processed real UMD course data by building **Python** scraping pipelines and **JSON APIs**, aggregating professor ratings, average GPA, and prerequisites through custom parsers for accurate, data-driven planning.

Gesture Control for Apple TV – OpenCV, MediaPipe, NumPy | [GitHub](#)

- Built a custom gesture recognition system by collecting and labeling a hand-gesture dataset, training a classifier on **MediaPipe** landmarks, and integrating it with **OpenCV** for real-time inference.
- Connected the recognition pipeline to Apple TV using **pyatv** and designed a **multithreaded** event loop to minimize latency and ensure consistent Wi-Fi command execution.

SKILLS

Technical: Git, Linux, Docker, CI/CD, ETL Pipelines, SQL, PostgreSQL, MongoDB, React, Node.js, AWS, GCP
Languages: English (Professional), Mandarin (Native)